

* Vectors: look at how to use them

$$\vec{r} = (x, y, z)$$

Real valued functions: $f: D \longrightarrow \mathbb{R}$
 $f(t)$

Vector-valued functions: $\vec{F}: D \longrightarrow \mathbb{R}^3$

$$\vec{F}(t) = (F_x(t), F_y(t), F_z(t)) = \underbrace{F_x(t)}_{\substack{\uparrow \\ \text{"Real-valued"}}} \hat{e}_x + F_y(t) \hat{e}_y + F_z(t) \hat{e}_z$$

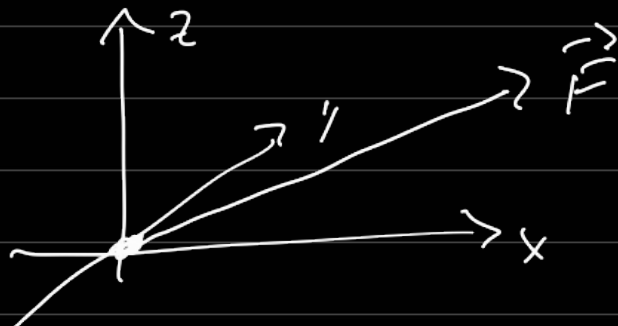
t : variable

• Vectors:

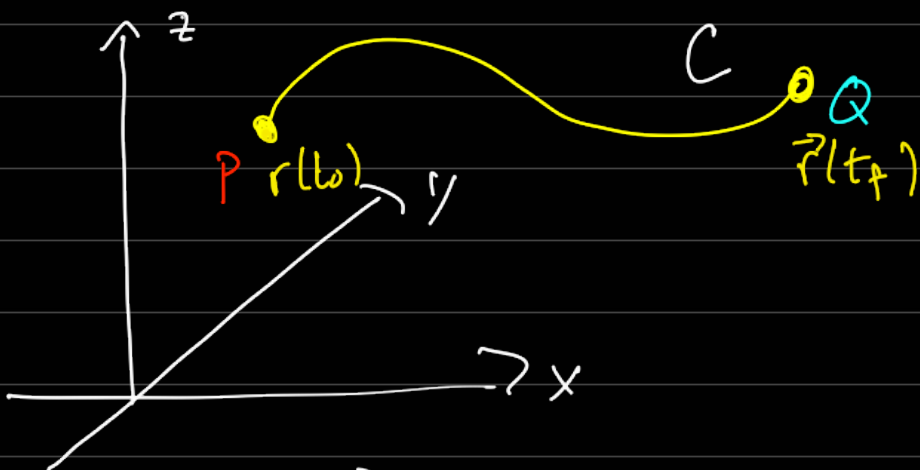
(i) mag

(ii) direction

Component functions



* Position vector: Parametrized curve



$$C(t) = \vec{r}(t) = (x(t), y(t), z(t))$$

some t_0 to t_f

$$\vec{r}(t_0) = P \quad | \quad \vec{r}(t_f) = Q$$

* Vec. Calculus: why?

• In basic physics we have

$$1D \quad \underline{F}_x = \frac{dp}{dt} = m \frac{d^2x}{dt^2} = ma$$

└

$\underline{F}_x$: (x-direction)

F: Force

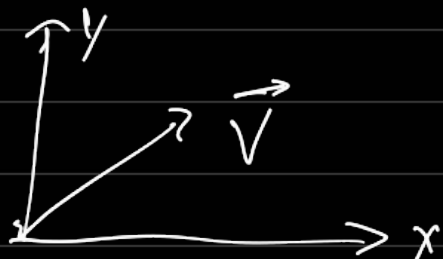
m: mass

x: position

$$\text{Velocity: } v = \frac{\Delta x}{\Delta t} \rightarrow \frac{dx}{dt}$$

→ Vector: • Speed: (magnitude of velocity)

• To each vel. we associate the mass to it



$$\text{momentum: } p_x = \underline{m} \underline{v}_x \quad | \quad \underline{v} = \frac{dx}{dt}$$

$$\text{Newton's laws: } \underline{F}_x = \frac{dp}{dt} = \frac{d(mv)}{dt} = \frac{dm}{dt} v + m \frac{dv}{dt}$$

generally, $\frac{dm}{dt} = 0$

→ Usually applied for looking at Rockets

• We'll assume $\frac{dm}{dt} = 0$

$$\underline{F}_x = m \frac{dv}{dt} = m \frac{d^2x}{dt^2}$$

acceleration: rate of change of velocity

* Want to look at these equations in 3D

→ Real-valued functions: $f: D \rightarrow \mathbb{R}$

$$\frac{df}{dx} = f'(x) = \lim_{\Delta x \rightarrow 0} \left(\frac{f(x + \Delta x) - f(x)}{\Delta x} \right)$$

Vector-valued: $\vec{F}(t)$

$$\frac{d\vec{F}}{dt} = \lim_{\Delta t \rightarrow 0} \left(\frac{\vec{F}(t + \Delta t) - \vec{F}(t)}{\Delta t} \right)$$

• Recall for vectors: $A = (A_x, A_y, A_z)$, $B = (B_x, B_y, B_z)$

$$\vec{A} + \vec{B} = (A_x + B_x, \dots)$$

$$\vec{F}(t + \Delta t) - \vec{F}(t) = (F_x(t + \Delta t) - F_x(t), F_y(t + \Delta t) - F_y(t), \dots)$$

$$\hookrightarrow \vec{F}(t + \Delta t) = (F_x(t + \Delta t), F_y(t + \Delta t), F_z(t + \Delta t))$$

• divide by Δt

$$\frac{d\vec{F}}{dt} = (F'_x(t), F'_y(t), F'_z(t)) = \underline{F'_x} \hat{e}_x + \underline{F'_y} \hat{e}_y + \underline{F'_z} \hat{e}_z$$

• Similarly: Integrals line-integrals

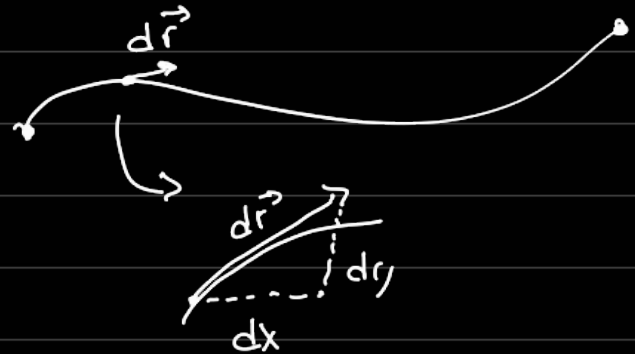
(i) $\int \phi(r) d\vec{r}$: Vector $\phi(r)$: function

(ii) $\int \vec{A} \cdot d\vec{r}$: Scalar

(iii) $\int \vec{A} \times d\vec{r}$: Vector

$$d\vec{r} = dx \hat{e}_x + dy \hat{e}_y + dz \hat{e}_z$$

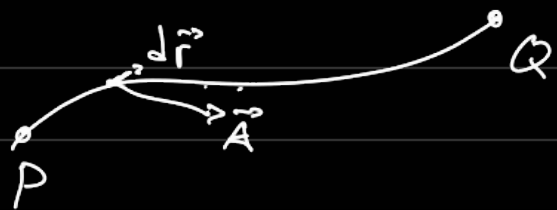
$$|d\vec{r}|^2 = dx^2 + dy^2 + dz^2$$



$$\begin{aligned} \text{(i)} \int \phi(r) d\vec{r} &= \int \phi(r) (dx \hat{e}_x + dy \hat{e}_y + dz \hat{e}_z) \\ &= \left(\int \phi(r) dx \right) \underline{\hat{e}_x} + \left(\int \phi(r) dy \right) \underline{\hat{e}_y} + \left(\int \phi(r) dz \right) \underline{\hat{e}_z} \end{aligned}$$

$$\text{(ii)} \int \vec{A} \cdot d\vec{r} = \int (A_x dx + A_y dy + A_z dz)$$

Arc-length: length of a curve

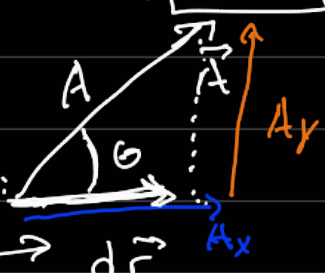


$$L \equiv \int_P^Q |d\vec{r}|$$

why?

$$\int \vec{A} \cdot d\vec{r}$$

$(\vec{A} \cdot d\vec{r})$: (component of $\vec{A}$ along the direction of $d\vec{r}$)



$$\vec{A} \cdot d\vec{r} = |A| |dr| \cos(\theta)$$

$$A_y / A_x = \tan(\theta)$$

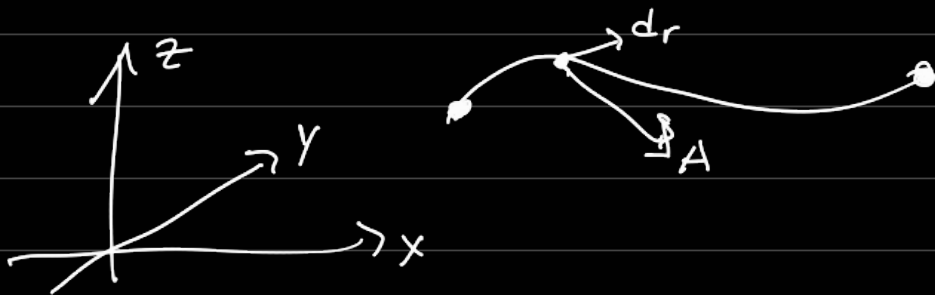
$$\cos(\theta) = \frac{A_x}{|A|}$$

$$A_x = |A| \cos(\theta)$$

$$\vec{A} \cdot d\vec{r} = A_x |dr|$$

→ Instead of $\vec{A} \cdot d\vec{r}$ we instead want $d\vec{r} \cdot d\vec{r} = |d\vec{r}|^2$.

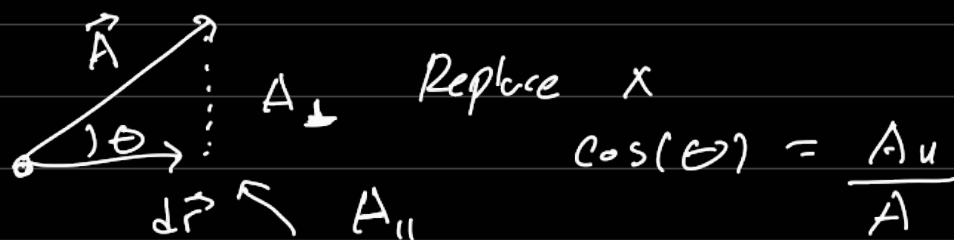
• In general



To make thing clear:

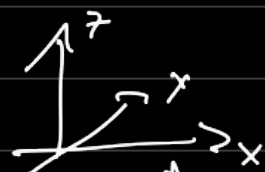
→ Set up system: $d\vec{r}$ was along x-direction
Not true in general

In general: $\vec{A} \cdot d\vec{r}$



$$\cos(\theta) = \frac{A_{||}}{A}$$

$$A_{||} = A \cos(\theta)$$

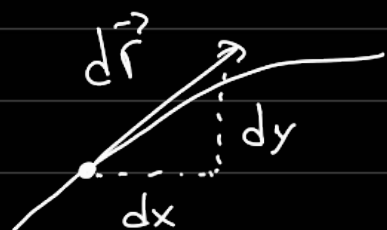


• In general: $d\vec{r}$: goes along same direction

$$d\vec{r} \cdot d\vec{r} = |d\vec{r}|^2 \Rightarrow |d\vec{r}| = \sqrt{d\vec{r} \cdot d\vec{r}}$$

Is close to what we want:

Are-length: $L = \int_{t_0}^t |d\vec{r}|$



$$\frac{|d\vec{r}|^2}{dt^2} = \frac{dx^2 + dy^2 + dz^2}{dt^2}$$

$$|d\vec{r}|^2 = |dx|^2 + |dy|^2 + |dz|^2$$

$$dt^2 \left(\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 + \left(\frac{dz}{dt} \right)^2 \right)$$

$$|d\vec{r}|^2 = dt^2 \left(\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 + \left(\frac{dz}{dt} \right)^2 \right)$$

Take $\sqrt{\quad}$:

$$|d\vec{r}| = dt \sqrt{(x')^2 + (y')^2 + (z')^2}$$

Plug back into L

$$L = \int |d\vec{r}| = \int_{t_0}^t \sqrt{(x')^2 + (y')^2 + (z')^2} dt$$

* Uni: differential forms

Back to Newton:

$\vec{F}$: Force vector

$$\vec{F}(t) = (F_x(t), F_y(t), F_z(t))$$

x, y, z : positions

$$\begin{aligned} \vec{F} &= m \frac{d\vec{r}}{dt} = m \left(\frac{d^2x(t)}{dt^2}, \frac{d^2y(t)}{dt^2}, \frac{d^2z(t)}{dt^2} \right) \\ &= (F_x(t), F_y(t), F_z(t)) \end{aligned}$$

(Vector equation)

• 3 equations in 1:

$$F_x(t) = m \frac{d^2x(t)}{dt^2}$$

$$F_y(t) = m \frac{d^2y(t)}{dt^2}$$

$$F_z(t) = m \frac{d^2z(t)}{dt^2}$$

In GR: tensors

$$G_{\mu\nu} = T_{\mu\nu}$$

~ 36 equations in 1

(Differential Geometry)

• QM: most is probability

